RESEARCH REPORT 3

SYSTEMS ENGINEERING FOR MANAGERS
49004

GROUP RED 4

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UTS: ENGINEERING FIELD – SYSTEMS ENGINEERING FOR MANAGERS
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High-Level Concept Design: Waste Management System

1. Solutions to Client's Requirements:

1.1. QR Codes:

Using existing space for disposal instructions, we can replace it for a QR Code with the disposal instructions within the code [1]. Above the QR code a click bait is suggested to encourage people to scan the QR code, e.g., "Scan me to win prizes!"

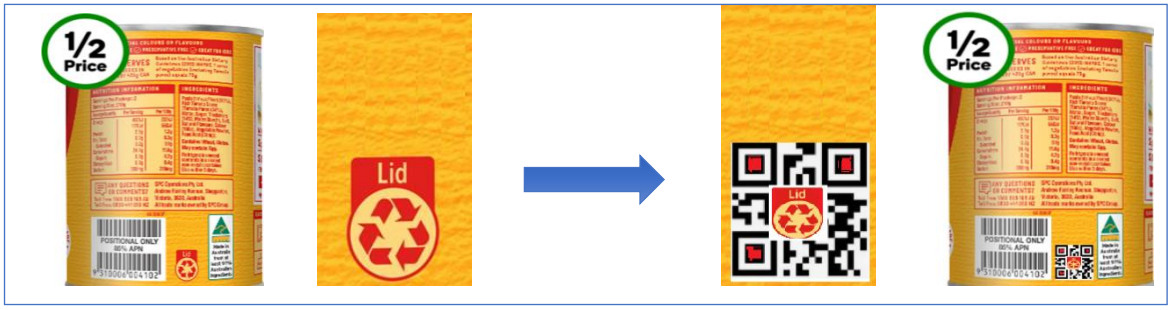

Figure 1: QR Codes in packaging

Once the QR code is scanned, the consumer will be re-directed to the APCO website [2], where the options for the correct disposal of the product and the waste management training will be given.

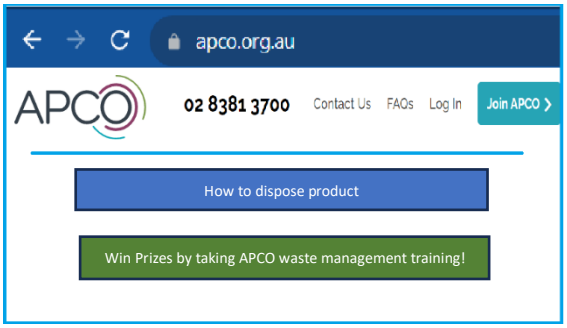

Figure 2. QR Code Link

1.2. Smart Bins:

The second phase of the project, we are suggesting smart bins for household use. These bins will be compatible with the QR code of the products.

Figure 3. Smart Bin

Scanning: Scan the product you want to dispose, the corresponding waste compartment door will open, then just drop the waste. The waste is now sorted.
Weight Sensor: If an item is significantly heavier than its standard weight (indicating its filled with liquid or has food residue) the bin would flag this as potentially contaminated [3][4].
Manual Disposal: For products with no QR code, the smart bin has an in-build screen that can guide you to correctly dispose the waste [5].

1.3. Smart Garbage Chute System:

The third phase of the project, we are implementing a smart chute system that is compatible with the smart bin. Once any of the smart bin compartments is full, the screen will indicate it. The bag of waste is removed, the smart bin will print a label that can be scanned by the smart chute. Once scanned, the chute's waste compartment door will open, the user can dispose of the bag of waste [6].

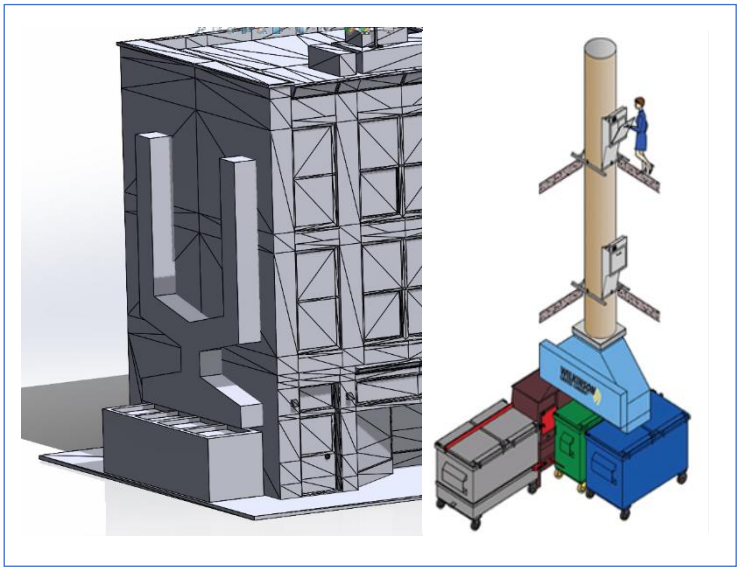

Figure 3: Smart Chute System

2. <u>Meeting System Requirements:</u>

2.1. Reduction of Cross Contamination:

By increasing the waste sorting efficiency and reducing the waste sorting error rate, the amount of cross contamination of waste is reduced significantly.

2.2. Reduction of waste generation from MUDs:

By completing the APCO waste management training, the people are become aware about the waste production, and they proactively try to reduce the waste they produce. This in turn reduces the waste generated by MUDs.

2.3. Designated waste bins:

Our smart bins have designated waste storage based on the waste category that increases MUD tenants' involvement in waste sorting and the smart chute delivery system reduces the MUD tenant's effort to sort waste.

2.4. Increased resource recovery:

The waste being efficiently sorted, it gives the existing system in place to recover the resources much better and improves the loss of resources.

2.5. Increase recyclable output:

Uncontaminated waste produces more recyclable products, and most resources are recovered. By introducing this new smart bins, contamination of waste is significantly reduced, increasing the recyclable output.

2.6. Collect more volume of waste per square area:

The smart chute creates an ingenious way of transporting the waste directly to a bin that's easily accessible by the garbage trucks thereby increasing the amount of waste collected per square area.

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